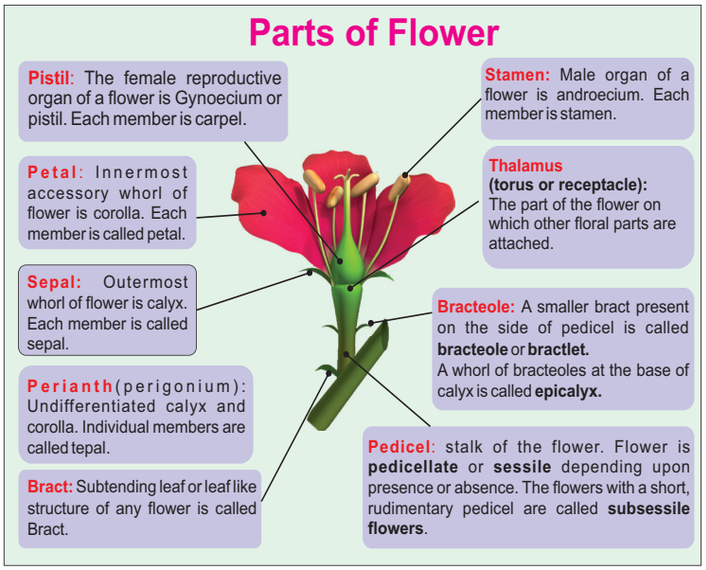


Figure 4.9: Parts of flower

2. Hypanthodium: Receptacle is a hollow, globose structure consisting of unisexual flowers present on the inner wall of the receptacle. Receptacle is closed leaving a small opening called **ostiole** which is covered by a series of bracts. Male flowers are present nearer to the ostiole, female and neutral flowers are found in a mixed manner from middle below. Example: *Ficus sp*. (Banyan, Fig and Pipal).

3. Coenanthium: Circular disc like fleshy open receptacle that bears pistillate flowers at the center and staminate flowers at the periphery. Example: *Dorstenia*

4.2 Flower

In a plant, which part would you like the most? Of course, it is a flower, because of its colour and fragrance. The flower is a significant diagnostic feature of angiosperms. It is a modified condensed reproductive shoot. The growth of the flower shoot is determinate.

4.2.1 Whorls of flower

There are two whorls, accessory and essential. Accessory whorl consists of calyx and corolla and essential whorl comprises of androecium and gynoecium.

Flower is said to be **Complete** when it contains all four whorls. An **Incomplete** flower is devoid of one or more whorls.

4.2.2 Flower sex

Flower sex refers to the presence or absence of androecium and gynoecium within a flower.

1. Perfect or bisexual: If a flower contains both androecium and gynoecium it is called as a **perfect flower**.

2. Imperfect or unisexual: When the flower contains only one of the essential whorls is called **Imperfect flower**. It is of two types:

i) **Staminate flowers**: Flowers with androecium alone.

ii) Pistillate flowers: Flowers with gynoecium alone.



Figure 4.10: (a) Bisexual flower

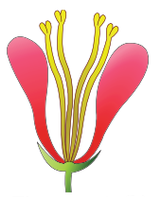


Figure 4.10: (b) Male flower

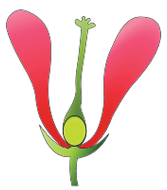


Figure 4.10: (c) Female flower

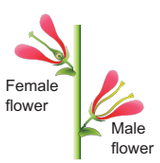


Figure 4.11: (a) Monoecious



Figure 4.11: (b) Dioecious



Figure 4.11: (c) Polygamous

4.2.3 Plant sex

Plant sex refers to the presence and distribution of flowers with different sexes in an individual plant.

1. **Hermaphroditic**: All the flowers of the plant are bisexual.

2. **Monoecious**: Both male and female flowers are present in the same plant Example: Coconut.

3. **Dioecious**: Male and Female flowers are present on separate plants. Example: Papaya, Palmyra.

4. **Polygamous**: The condition in which bisexual and unisexual (staminate/pistillate) flowers occur in a same plant is called **polygamous**. Example: *Musa*, *Mangifera*.

11th Botany_EM_Chapter 4.indd 84 05-01-2023 18:22:40